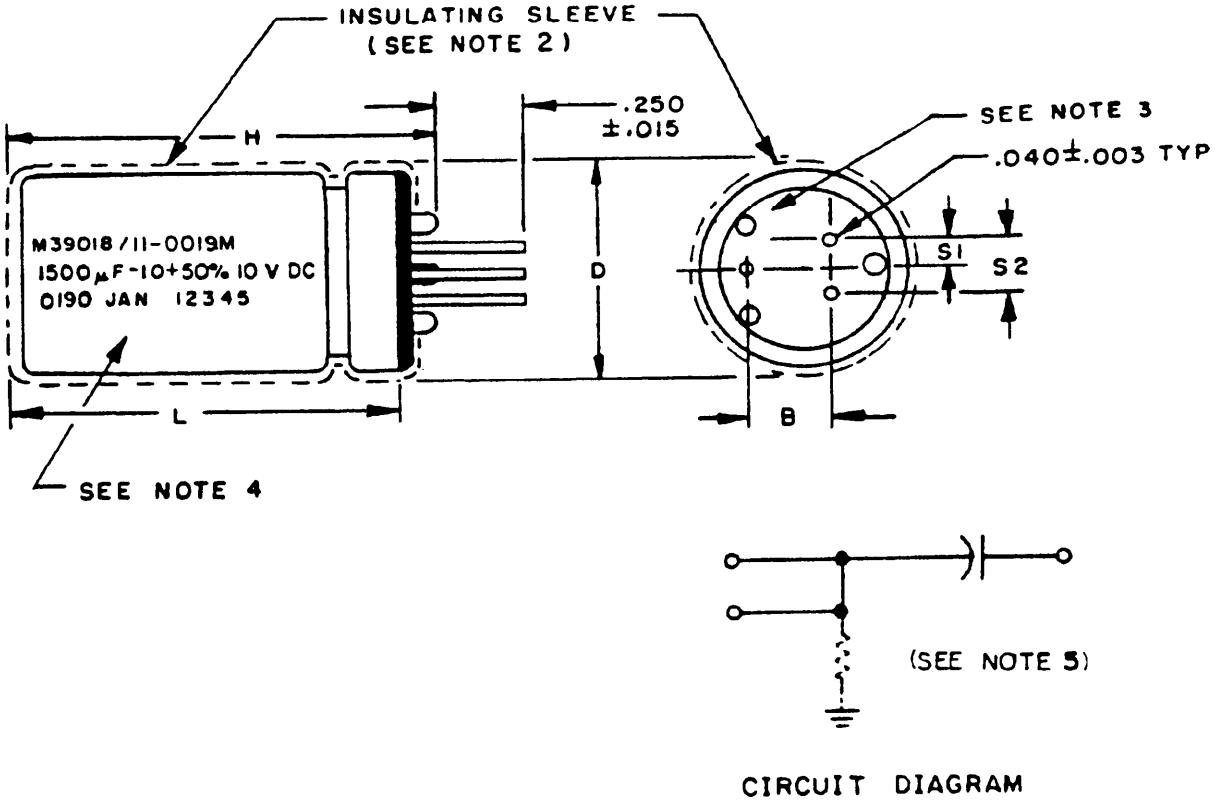
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (ALUMINUM OXIDE) (POLARIZED), ESTABLISHED RELIABILITY, STYLES CUR03 (INSULATED) AND CUR04 (INSULATED)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-39018.



CIRCUIT DIAGRAM

Style CUR03

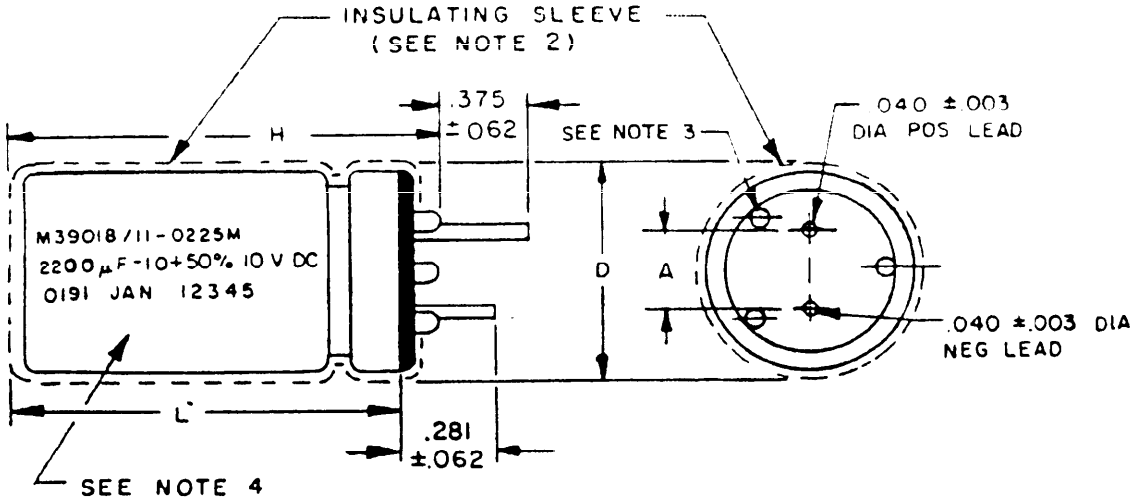
FIGURE 1. Styles CUR03 and CUR04 insulated capacitors.

AMSC N/A

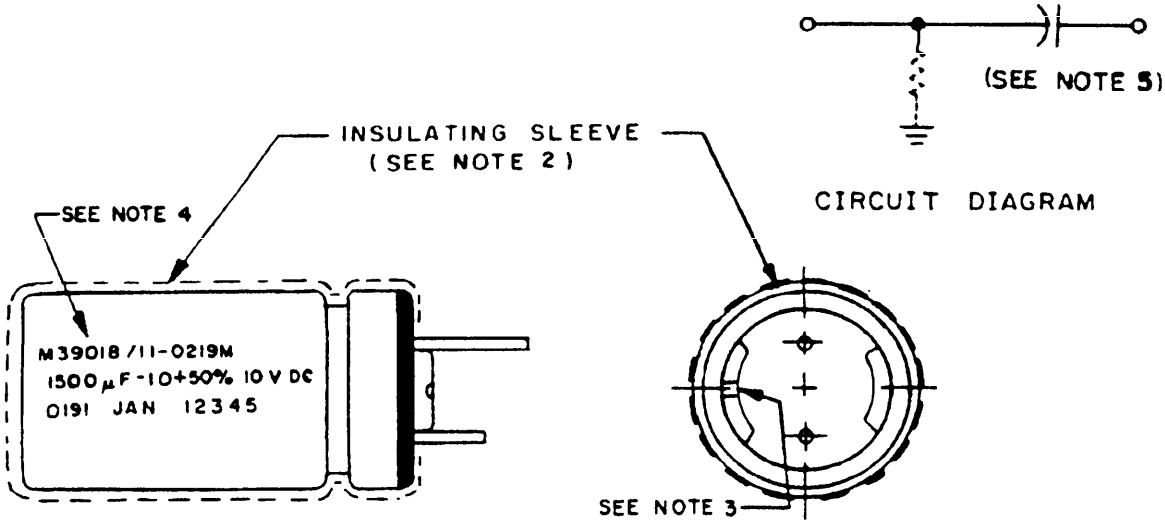
FSC 5910

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Case sizes A1 through F1, and A2 through F2.



Case sizes A0 through F0 (see note 6).

STYLE CUR04

FIGURE 1. Styles CUR03 and CUR04 insulated capacitors - Continued.

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Dimensions							
Case size	L ±.062	D ±.015	H max	A ±.015	B ±.015	S1 ±.015	S2 ±.015
A0	1.125	.750	1.246	.250	.300	.100	.200
B0	1.625	.750	1.746	.250	.300	.100	.200
C0	2.125	.750	2.246	.250	.300	.100	.200
D0	2.625	.750	2.746	.250	.300	.100	.200
E0	3.125	.750	3.246	.250	.300	.100	.200
F0	3.625	.750	3.746	.250	.300	.100	.200
A1	1.125	.875	1.246	.300	.400	.150	.300
B1	1.625	.875	1.746	.300	.400	.150	.300
C1	2.125	.875	2.246	.300	.400	.150	.300
D1	2.625	.875	2.746	.300	.400	.150	.300
E1	3.125	.875	3.246	.300	.400	.150	.300
F1	3.625	.875	3.746	.300	.400	.150	.300
A2	1.125	1.000	1.246	.400	.400	.150	.300
B2	1.625	1.000	1.746	.400	.400	.150	.300
C2	2.125	1.000	2.246	.400	.400	.150	.300
D2	2.625	1.000	2.746	.400	.400	.150	.300
E2	3.125	1.000	3.246	.400	.400	.150	.300
F2	3.625	1.000	3.746	.400	.400	.150	.300

Inches	mm
.003	0.08
.015	0.38
.040	1.02
.062	1.57
.100	2.54
.150	3.81
.200	5.08
.250	6.35
.281	7.14
.300	7.62
.375	9.52
.400	10.16
.750	19.05
.875	22.22
1.000	25.40
1.125	28.58
1.246	31.65
1.625	41.28
1.746	44.35
2.125	53.98
2.246	57.05
2.625	66.68
2.746	69.75
3.125	79.38
3.246	82.45
3.625	92.08
3.746	95.15

NOTES:
1. Dimensions are in inches.
2. Insulating sleeve thickness shall not exceed .008 (0.20 mm) and shall overlap the ends of the capacitor body.
3. Standoff shape is optional. The minimum height of the standoffs shall be 0.025 (0.64 mm).
4. Example of marking and its relative location on the capacitor body.
5. There is an indeterminate resistance between the metal case and the negative terminals.
6. All dimensions shown on the figure for CUR04 case sizes A1 through F1 and A2 through F2 also apply for case sizes A0 through F0.
7. Metric equivalent are given for general information only.

FIGURE 1. Styles CUR03 and CUR04 insulated capacitors - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case: Type - Tubular
Material - Metal.

Terminals: See figure 1.

DC rated voltage: See table I.

Reverse voltage: 1.5 volts maximum.

Capacitance (Cap) value: See table I.

Cap tolerance: -10, +50%.

Failure rate level: M, P, R, and S.

Operating temperature range: -55°C to 105°C.

DC leakage (DCL): See table I.

Equivalent series resistance (ESR): See table I.

Low temperature exposure:
DCL - See table I.
Cap - See table I.
ESR - See table I.

Terminal strength: Method 211, MIL-STD-202, test conditions A (3 pounds) and D (3 rotations).

Stability at reduced and high temperatures:

Step 1 (at +25°C):
DCL - See table I.
Cap - See table I.
ESR - See table I.

Step 2 (at -55°C):
Impedance - See table I.
Cap - Within the following percentages of the step 1 value:
7.5 V dc: ±35 percent.
10 to 30 V dc, inclusive: ±25 percent.
50 to 100 V dc, inclusive: ±20 percent.
150 to 250 V dc, inclusive: ±30 percent.

Step 3 (at +25°C):
DCL - See table I.
Cap - See table I.
ESR - See table I.

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TABLE I. Styles CUR[illegible]d CUR04 capacitors.

DC rated voltage at +105°C	Nominal capacitance	DC surge voltage at +105°C	ESR at 25°C		Maximum impedance at 100 kHz		Maximum dc leakage		Maximum ripple current at +85°C (20 k - 100 kHz)	Case size	Dash number 1/	
			120 Hz max	20 kHz-100 kHz max	+25°C	-55°C	+25°C	+105°C			CUR03	CUR04
Volts	µF	Volts	Milliohms		Milliohms		µA		Amperes			
7.5	1800	10	103.0	73.0	77	616	58	348	2.41	A0	0001-	0201-
"	3900	"	52.0	38.0	41	328	86	516	3.86	B0	0002-	0202-
"	5600	"	36.0	27.0	31	248	102	612	5.12	C0	0003-	0203-
"	6800	"	28.0	22.0	27	216	112	672	6.21	D0	0004-	0204-
"	8200	"	24.0	18.0	23	184	124	744	7.41	E0	0005-	0205-
"	12000	"	20.5	16.0	21	168	150	900	8.42	F0	0006-	0206-
7.5	2700	10	67.0	47.0	51	408	71	426	3.31	A1	0007-	0207-
"	5600	"	34.0	24.0	28	224	102	612	5.32	B1	0008-	0208-
"	8200	"	23.5	17.0	22	176	124	744	7.05	C1	0009-	0209-
"	12000	"	18.5	13.0	19	152	150	900	8.81	D1	0010-	0210-
"	15000	"	15.0	11.0	18	144	168	1008	10.30	E1	0011-	0211-
"	18000	"	13.0	9.5	17	132	184	1104	11.90	F1	0012-	0212-
7.5	3900	10	50.0	35.0	39	312	86	516	4.18	A2	0013-	0213-
"	8200	"	26.0	18.2	21	184	124	744	6.64	B2	0014-	0214-
"	12000	"	18.0	13.0	19	152	150	900	8.72	C2	0015-	0215-
"	15000	"	14.0	10.4	18	144	168	1008	10.60	D2	0016-	0216-
"	18000	"	12.0	8.6	16	128	184	1104	12.60	E2	0017-	0217-
"	22000	"	10.4	7.5	15	120	203	1218	14.40	F2	0018-	0218-
10	1500	15	111.0	73.0	75	600	61	366	2.41	A0	0019-	0219-
"	3300	"	56.0	38.0	41	328	91	546	3.86	B0	0020-	0220-
"	4700	"	39.0	27.0	31	248	108	648	5.12	C0	0021-	0221-
"	6800	"	30.0	21.0	25	200	130	780	6.35	D0	0022-	0222-
"	8200	"	25.0	18.0	22	176	143	858	7.40	E0	0023-	0223-
"	10000	"	22.0	16.0	21	168	158	948	8.42	F0	0024-	0224-
10	2200	15	72.0	47.0	50	400	74	444	3.31	A1	0025-	0225-
"	4700	"	37.0	24.0	28	224	108	648	5.32	B1	0026-	0226-
"	6800	"	25.0	17.0	22	176	130	780	7.05	C1	0027-	0227-
"	8200	"	18.0	13.0	19	152	143	858	8.81	D1	0028-	0228-
"	12000	"	16.0	11.0	17	136	173	1038	10.30	E1	0029-	0229-
"	15000	"	14.0	9.4	15	120	194	1164	12.00	F1	0030-	0230-
10	3300	15	54.0	35.0	39	312	91	546	4.18	A2	0031-	0231-
"	6800	"	28.0	10.5	23	184	130	780	6.58	B2	0032-	0232-
"	10000	"	19.0	13.0	19	152	158	948	8.72	C2	0033-	0233-
"	12000	"	15.0	10.2	18	144	173	1038	10.80	D2	0034-	0234-
"	15000	"	13.0	8.6	16	128	194	1164	12.60	E2	0035-	0235-
"	22000	"	11.0	7.6	15	120	235	1410	14.30	F2	0036-	0236-

See footnote at end of table.

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TABLE I. Styles CUR03 and CUR04 capacitors - Continued.

DC rated voltage at +105°C	Nominal capacitance	DC surge voltage at +105°C	ESR at 25°C		Maximum impedance at 100 kHz		Maximum dc leakage		Maximum ripple current at +85°C (20 k - 100 kHz)	Case size	Dash number 1/	
			120 Hz max	20 kHz-100 kHz max	+25°C	-55°C	+25°C	+105°C			CUR03	CUR04
Volts	µF	Volts	Milliohms		Milliohms		µA		Amperes			
16	1200	20	114.0	73.0	75	600	69	414	2.41	A0	0037-	0237-
"	2200	"	57.0	38.0	41	328	94	564	3.86	B0	0038-	0238-
"	3300	"	40.0	27.0	30	240	115	690	5.12	C0	0039-	0239-
"	4700	"	31.0	21.0	24	192	137	822	6.35	D0	0040-	0240-
"	5600	"	26.0	18.0	22	176	150	900	7.40	E0	0041-	0241-
"	6800	"	22.0	16.0	21	168	165	990	8.42	F0	0042-	0242-
16	1800	20	74.0	47.0	50	400	85	510	3.31	A1	0043-	0243-
"	3300	"	38.0	24.0	28	224	115	690	5.32	B1	0044-	0244-
"	5600	"	26.0	17.0	21	168	150	900	7.05	C1	0045-	0245-
"	6800	"	20.0	13.0	19	152	165	990	8.81	D1	0046-	0246-
"	8200	"	16.6	11.0	17	136	181	1086	10.30	E1	0047-	0247-
"	10000	"	14.3	9.5	16	128	200	1200	11.90	F1	0048-	0248-
16	2200	20	55.0	35.0	38	304	94	564	4.18	A2	0049-	0249-
"	4700	"	28.0	18.5	23	184	137	822	6.58	B2	0050-	0250-
"	6800	"	20.0	13.0	19	152	165	990	8.72	C2	0051-	0251-
"	10000	"	16.0	10.3	17	136	200	1200	10.70	D2	0052-	0252-
"	12000	"	13.0	8.6	16	128	219	1314	12.60	E2	0053-	0253-
"	15000	"	11.5	7.5	15	120	245	1470	14.40	F2	0054-	0254-
30	560	40	162.0	74.0	76	760	65	390	2.40	A0	0055-	0255-
"	1200	"	81.0	38.0	41	410	95	570	3.86	B0	0056-	0256-
"	1800	"	55.0	27.0	31	310	116	696	5.11	C0	0057-	0257-
"	2200	"	42.0	21.0	25	250	128	768	6.36	D0	0058-	0258-
"	2700	"	35.0	18.0	23	230	142	852	7.40	E0	0059-	0259-
"	3300	"	30.0	15.7	19	190	157	942	8.50	F0	0060-	0260-
30	820	40	107.0	47.0	50	500	78	468	3.31	A1	0061-	0261-
"	1800	"	54.0	24.0	28	280	116	696	5.32	B1	0062-	0262-
"	2700	"	36.0	16.8	22	220	142	852	7.10	C1	0063-	0263-
"	3300	"	28.0	13.0	18	180	157	942	8.81	D1	0064-	0264-
"	3900	"	23.0	11.0	16	160	171	1026	10.30	E1	0065-	0265-
"	4700	"	19.5	9.4	14	140	188	1128	12.00	F1	0066-	0266-
30	1200	40	78.0	36.0	38	380	95	570	4.12	A2	0067-	0267-
"	2200	"	39.0	18.5	23	230	128	768	6.58	B2	0068-	0268-
"	3900	"	27.0	13.0	19	190	171	1026	8.72	C2	0069-	0269-
"	4700	"	21.0	10.3	17	170	188	1128	10.70	D2	0070-	0270-
"	5600	"	17.0	8.6	16	160	205	1230	12.60	E2	0071-	0271-
"	6800	"	14.8	7.5	15	150	226	1356	14.40	F2	0072-	0272-

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TABLE I. Styles CUR03 and CUR04 capacitors - continued.

DC rated voltage at +105°C	Nominal capacitance	DC surge voltage at +105°C	ESR at 25°C		Maximum impedance at 100 kHz		Maximum dc leakage		Maximum ripple current at +85°C (20 k - 100 kHz)	Case size	Dash number 1/	
			120 Hz max	20 kHz-100 kHz max	+25°C	-55°C	+25°C	+105°C			CUR03	CUR04
Volts	µF	Volts	Milliohms		Milliohms		µA		Amperes			
50	270	75	255.0	74.0	76	1140	58	348	2.40	A0	0073-	0273-
"	560	"	125.0	39.0	41	615	84	504	3.81	B0	0074-	0274-
"	820	"	84.0	27.0	31	465	101	606	5.12	C0	0075-	0275-
"	1200	"	64.0	22.0	26	390	122	732	6.21	D0	0076-	0276-
"	1500	"	52.0	18.0	22	330	137	822	7.40	E0	0077-	0277-
"	1800	"	44.0	15.0	20	300	150	900	8.69	F0	0078-	0278-
50	470	75	169.0	47.0	50	750	77	462	3.31	A1	0079-	0279-
"	820	"	83.0	24.0	27	405	101	606	5.32	B1	0080-	0280-
"	1200	"	56.0	17.0	23	345	122	732	7.05	C1	0081-	0281-
"	1800	"	43.0	13.0	19	285	150	900	8.81	D1	0082-	0282-
"	2200	"	35.0	11.0	17	255	166	996	10.30	E1	0083-	0283-
"	2700	"	29.0	9.5	16	240	184	1104	11.90	F1	0084-	0284-
50	560	75	121.0	36.0	39	585	84	504	4.12	A2	0085-	0285-
"	1200	"	60.0	18.5	23	345	122	732	6.58	B2	0086-	0286-
"	1800	"	41.0	13.0	19	285	150	900	8.72	C2	0087-	0287-
"	2700	"	31.0	10.4	17	255	184	1104	10.60	D2	0088-	0288-
"	3300	"	26.0	8.6	16	240	203	1218	12.60	E2	0089-	0289-
"	3900	"	22.0	7.6	15	225	221	1326	14.30	F2	0090-	0290-
63	220	85	315.0	75.0	77	1155	59	354	2.38	A0	0091-	0291-
"	390	"	154.0	39.0	41	615	78	468	3.81	B0	0092-	0292-
"	680	"	104.0	27.0	31	465	104	624	5.12	C0	0093-	0293-
"	820	"	79.0	22.0	26	390	114	684	6.21	D0	0094-	0294-
"	1000	"	64.0	18.0	22	330	125	750	7.40	E0	0095-	0295-
"	1200	"	54.0	16.0	20	300	137	822	8.42	F0	0096-	0296-
63	330	85	210.0	48.0	51	765	72	432	3.28	A1	0097-	0297-
"	680	"	103.0	24.4	28	420	104	624	5.28	B1	0098-	0298-
"	1000	"	69.0	17.0	23	345	125	750	7.05	C1	0099-	0299-
"	1200	"	52.0	13.0	19	285	137	822	8.81	D1	0100-	0300-
"	1500	"	42.0	11.0	17	255	154	924	10.30	E1	0101-	0301-
"	1800	"	36.0	9.5	16	240	168	1008	11.90	F1	0102-	0302-
63	480	85	149.0	36.0	39	585	87	522	4.12	A2	0103-	0303-
"	820	"	74.0	18.5	23	345	114	684	6.58	B2	0104-	0304-
"	1200	"	50.0	13.0	19	285	137	822	8.72	C2	0105-	0305-
"	1800	"	38.0	10.4	17	255	168	1008	10.60	D2	0106-	0306-
"	2200	"	31.0	8.7	16	240	186	1116	12.50	E2	0107-	0307-
"	2700	"	26.0	7.6	15	225	206	1236	14.30	F2	0108-	0308-

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TABLE I. Styles CUR03 and CUR04 capacitors - Continued.

DC rated voltage at +105°C	Nominal capacitance	DC surge voltage at +105°C	ESR at 25°C		Maximum impedance at 100 kHz		Maximum dc leakage		Maximum ripple current at +85°C (20 k - 100 kHz)	Case size	Dash number 1/	
			120 Hz max	20 kHz-100 kHz max	+25°C	-55°C	+25°C	+105°C			CUR03	CUR04
Volts	µF	Volts	Milliohms		Milliohms		µA		Amperes			
75	100	100	567.0	107.0	118	2360	44	264	2.00	A0	0198-	0398-
"	220	"	280.0	54.0	59	1180	64	384	3.23	B0	0109-	0309-
"	330	"	185.0	38.0	44	880	79	474	4.31	C0	0110-	0310-
"	390	"	140.0	29.0	35	700	86	516	5.41	D0	0111-	0311-
"	560	"	112.0	24.0	31	620	102	612	6.41	E0	0112-	0312-
"	680	"	94.0	21.0	29	580	112	672	7.35	F0	0113-	0313-
75	180	100	380.0	77.0	81	1620	58	348	2.59	A1	0114-	0314-
"	330	"	186.0	40.0	44	880	79	474	4.12	B1	0115-	0315-
"	470	"	125.0	28.0	33	660	94	564	5.50	C1	0116-	0316-
"	680	"	94.0	22.0	29	580	112	672	6.78	D1	0117-	0317-
"	820	"	76.0	19.0	25	500	124	744	7.87	E1	0118-	0318-
"	1000	"	64.0	16.0	23	460	137	822	9.17	F1	0119-	0319-
75	220	100	278.0	62.0	65	1300	64	384	3.14	A2	0120-	0320-
"	470	"	137.0	33.0	37	740	94	564	4.93	B2	0121-	0321-
"	680	"	92.0	23.0	28	560	112	672	6.56	C2	0122-	0322-
"	1000	"	70.0	18.6	24	480	137	822	7.96	D2	0123-	0323-
"	1200	"	57.0	16.0	23	460	150	900	9.25	E2	0124-	0324-
"	1500	"	48.0	14.0	21	420	168	1008	10.60	F2	0125-	0325-
100	82	125	1040.0	295.0	308	6160	45	270	1.20	A0	0126-	0326-
"	150	"	505.0	145.0	159	3180	61	366	1.97	B0	0127-	0327-
"	220	"	320.0	98.0	104	2080	74	444	2.69	C0	0128-	0328-
"	330	"	236.0	74.0	80	1600	91	546	3.39	D0	0129-	0329-
"	390	"	201.0	60.0	66	1320	99	594	4.06	E0	0130-	0330-
"	470	"	160.0	51.0	57	1140	108	648	4.71	F0	0131-	0331-
100	120	125	650.0	200.0	210	4200	55	330	1.61	A1	0132-	0332-
"	220	"	315.0	99.0	105	2100	74	444	2.62	B1	0133-	0333-
"	330	"	215.0	67.0	69	1380	91	546	3.55	C1	0134-	0334-
"	470	"	160.0	52.0	56	1120	108	648	4.41	D1	0135-	0335-
"	560	"	126.0	42.0	47	940	118	708	5.29	E1	0136-	0336-
"	680	"	107.0	36.0	40	800	130	780	6.11	F1	0137-	0337-
100	180	125	465.0	150.0	154	3080	67	402	2.02	A2	0138-	0338-
"	330	"	230.0	75.0	79	1580	91	546	3.27	B2	0139-	0339-
"	470	"	155.0	51.0	55	1100	108	648	4.40	C2	0140-	0340-
"	680	"	115.0	40.0	44	880	130	780	5.43	D2	0141-	0341-
"	820	"	94.0	33.0	37	740	143	858	6.44	E2	0142-	0342-
"	1000	"	78.0	28.0	32	640	158	948	7.46	F2	0143-	0343-

See footnote at end of table.

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TABLE I. Styles CUR03 and CUR0' capacitors - Continued.

DC rated voltage at +105°C	Nominal capacitance	DC surge voltage at +105°C	ESR at 25°C		Maximum impedance at 100 kHz		Maximum dc leakage		Maximum ripple current at +85°C (20 k - 100 kHz)	Case size	Dash number 1/	
			120 Hz max	20 kHz-100 kHz max	+25°C	-55°C	+25°C	+105°C			CUR03	CUR04
Volts	µF	Volts	Milliohms		Milliohms		µA		Amperes			
150	39	175	1980.0	590.0	605	15125	39	234	.850	A0	0144-	0344-
"	82	"	965.0	290.0	305	7625	56	336	1.400	B0	0145-	0345-
"	120	"	640.0	195.0	210	5250	67	402	1.900	C0	0146-	0346-
"	150	"	480.0	145.0	155	3875	75	450	2.420	D0	0147-	0347-
"	180	"	385.0	116.0	126	3150	82	492	2.920	E0	0148-	0348-
"	220	"	316.0	97.0	104	2600	91	546	3.420	F0	0149-	0349-
150	56	175	1370.0	410.0	425	10625	46	276	1.120	A1	0150-	0350-
"	120	"	665.0	205.0	214	5350	67	402	1.820	B1	0151-	0351-
"	180	"	440.0	135.0	143	3575	82	492	2.500	C1	0152-	0352-
"	220	"	330.0	101.0	109	2725	91	546	3.160	D1	0153-	0353-
"	270	"	265.0	82.0	88	2200	101	606	3.790	E1	0154-	0354-
"	330	"	220.0	69.0	76	1900	111	666	4.410	F1	0155-	0355-
150	82	175	975.0	297.0	309	7725	56	336	1.440	A2	0156-	0356-
"	180	"	475.0	146.0	152	3800	82	492	2.340	B2	0157-	0357-
"	270	"	315.0	98.0	104	2600	101	606	3.180	C2	0158-	0358-
"	330	"	236.0	75.0	81	2025	111	666	3.960	D2	0159-	0359-
"	390	"	189.0	61.0	68	1700	121	726	4.740	E2	0160-	0360-
"	470	"	159.0	51.0	58	1450	133	798	5.530	F2	0161-	0361-
200	33	225	2310.0	617.0	630	15750	41	246	.830	A0	0162-	0362-
"	68	"	1120.0	267.0	302	7550	59	354	1.400	B0	0163-	0363-
"	100	"	741.0	191.0	201	5025	71	426	1.920	C0	0164-	0364-
"	120	"	554.0	144.0	154	3850	78	468	2.430	D0	0165-	0365-
"	150	"	442.0	116.0	125	3125	87	522	2.920	E0	0166-	0366-
"	180	"	369.0	97.0	106	2650	95	570	3.420	F0	0167-	0367-
200	47	225	1600.0	411.0	422	10550	49	294	1.120	A1	0168-	0368-
"	82	"	776.0	202.0	212	5300	64	384	1.830	B1	0169-	0369-
"	150	"	514.0	134.0	143	3575	87	522	2.510	C1	0170-	0370-
"	180	"	384.0	102.0	111	2775	95	570	3.150	D1	0171-	0371-
"	220	"	308.0	82.0	91	2275	105	630	3.790	E1	0172-	0372-
"	270	"	257.0	69.0	78	1950	116	696	4.410	F1	0173-	0373-
200	68	225	1138.0	298.0	307	7675	59	354	1.430	A2	0174-	0374-
"	120	"	553.0	147.0	156	3900	78	468	2.340	B2	0175-	0375-
"	180	"	367.0	99.0	107	2675	95	570	2.590	C2	0176-	0376-
"	270	"	275.0	75.0	84	2100	116	696	3.960	D2	0177-	0377-
"	330	"	220.0	61.0	68	1700	129	774	4.740	E2	0178-	0378-
"	390	"	184.0	51.0	58	1450	140	840	5.530	F2	0179-	0379-

e footnote at end of table.

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TABLE I. Styles CUR03 and CUR04 capacitors - Continued.

DC rated voltage at +105°C	Nominal capacitance	DC surge voltage at +105°C	ESR at 25°C		Maximum impedance at 100 kHz		Maximum dc leakage		Maximum ripple current at +85°C (20 k - 100 kHz)	Case size	Dash number 1/	
			120 Hz max	20 kHz-100 kHz max	+25°C	-55°C	+25°C	+105°C			CUR03	CUR04
Volts	µF	Volts	Milliohms		Milliohms		µA		Amperes			
50	27	275	2790	590	602	15050	41	246	.850	A0	0180-	0380-
"	56	"	1350	290	301	7526	59	354	1.400	B0	0181-	0381-
"	82	"	895	192	202	5050	72	432	1.920	C0	0182-	0382-
"	100	"	669	145	155	3875	79	474	2.420	D0	0183-	0383-
"	120	"	535	116	125	3125	87	522	2.920	E0	0184-	0384-
"	150	"	445	98	107	2675	97	582	3.400	F0	0185-	0385-
50	39	275	1932	415	426	10650	50	300	1.120	A1	0186-	0386-
"	82	"	940	202	212	5300	72	432	1.840	B1	0187-	0387-
"	120	"	620	135	144	3600	87	522	2.500	C1	0188-	0388-
"	150	"	464	102	111	2775	97	582	3.150	D1	0189-	0389-
"	180	"	371	83	92	2300	106	636	3.760	E1	0190-	0390-
"	220	"	309	70	79	1975	117	702	4.380	F1	0191-	0391-
50	56	275	1370	300	309	7725	59	354	1.430	A2	0192-	0392-
"	120	"	668	148	157	2925	87	522	2.330	B2	0193-	0393-
"	180	"	442	99	107	2675	106	636	3.160	C2	0194-	0394-
"	220	"	331	75	84	2100	117	702	3.960	D2	0195-	0395-
"	270	"	265	61	68	1700	130	780	4.740	E2	0196-	0396-
"	330	"	221	52	58	1450	144	864	5.480	F2	0197-	0397-

Complete Part or Identifying Number (PIN) shall include a letter symbol to indicate failure rate level (i.e., 0001M, 0026R, etc).

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Step 4 (at +105°C):
DCL - See table I.
Cap - Shall not exceed ±30 percent of step 1 value.

Step 5 (at +25°C):
DCL - See table I.
Cap - Shall not exceed ±5 percent of step 1 value.
ESR - (at 120 Hz) see table I.

Life: Method 108, MIL-STD-202.
For qualification: Condition F (2,000 hours) at 85°C.
DCL - See table I.
Cap - Shall not exceed ±15 percent of initial measured value.
ESR - Not more than 150 percent of initial requirement.

10,000 hour (extended life and group C) at 85°C:
DCL - At 25°C, initial limit; at 85°C, 125 percent of initial limit.
Cap - Shall not exceed ±25 percent of initial measured value.
ESR - Not more than 200 percent of initial requirement.

Vibration:

High frequency - Method 204, MIL-STD-202, test condition A.

Random (style CUR04 only) - In accordance with MIL-C-39018.

Thermal shock and immersion:
Thermal shock - Method 107, MIL-STD-202, condition A, except in step 3, units shall be tested at applicable high temperature.

Immersion - Method 104, MIL-STD-202, condition B (2 cycles, 30 min.).
DCL - See table I.
Cap - Shall not exceed ±5 percent of initial measured value.
ESR - See table I.
Case insulation: Shall meet initial requirements.

Surge voltage:
DCL - See table I.
Cap - Shall not exceed ±12 percent of initial measured value.
ESR - See table I.

Moisture resistance - Method 106, MIL-STD-202.
DCL - See table I.
Cap - Shall not exceed ±6 percent of initial measured value.
ESR - See table I.
Case insulation: Shall meet initial requirements.

High temperature exposure (500 hours at 105°C): In accordance with MIL-C-39018.
DCL - Not more than 200 percent of initial requirement.
Cap - Shall not exceed ±10 percent of initial measured value.
ESR - Not more than 115 percent of initial requirement.

Penetration of solvents: In accordance with MIL-C-39018.

High temperature verification: In accordance with MIL-C-39018, at 105°C with rated voltage applied.

DCL - See table I.
Cap - Shall not exceed ±15 percent of initial measured value.
ESR - Not more than 150 percent of initial requirement.

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Reverse voltage aging: In accordance with MIL-C-39018.

Marking: In accordance with MIL-C-39018.

PIN: M39018/11- (dash number from table I).

Application notes:

1. Ripple current. Capacitors shall withstand the rms ripple current at 20 kHz through 100 kHz as given in table I.

   The maximum ripple current values apply with the following qualifications:
   The sum of the applied dc voltage plus the peak value of the impressed ac voltage must not exceed the dc rated voltage of the capacitor.

   The sum of the applied dc voltage plus the peak value of the impressed ac voltage must be such that not more than 1.5 volts reversal is experienced by the capacitor.

   If capacitors are operated at a frequency of less than 20 kHz, the 20 kHz rms ripple current must be multiplied by the factors in table II.

TABLE II. Factors for determining ripple current at various frequencies.

Rated voltage	Frequency (in hertz)				
	60	120	400	1K - 19K	20K - 100K
7.5-75	.60	.75	.80	.90	1.0
100-250	.43	.54	.75	.85	1.0

Where capacitors are operated at a temperature other than +85°C, the permissible rms ripple current must be multiplied by the factors in table III.

TABLE III. Factors for determining ripple current at various temperatures.

Ambient temperature	Factor
+105°C	0.3
+85	1.0
+65	1.4
+45	1.7
+25	2.0

When capacitors are operated at both a temperature and frequency other than 85°C and less than 20 kHz, respectively, both the multiplying factors described in tables II and III must be used to obtain the permissible rms ripple current.

Under no conditions of temperature or frequency, regardless of the multipliers given above, should the rms ripple current exceed 20 amperes.

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2. Insulation and grounding. These capacitors possess an indeterminate insulation resistance between the cathode terminal and the case. Where a potential difference exists between the cathode terminal and the chassis or other metallic mounting surfaces (which are usually at ground potential), the capacitors should be mounted in such a manner as to insulate the container from such mounting surfaces.

3. Operating life. These capacitors are expected to have a useful operating life in excess of ten years when subjected to typical circuit environments and operating conditions. Life can be extended by derating with regard to environmental and electrical requirements.

4. Shelf life. Shelf life and reforming procedures shall be in accordance with MIL-STD-1311.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Navy - SH
Air Force - 17, 99
DLA - ES

User activities:
Navy - AS, CG, MC, OS
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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